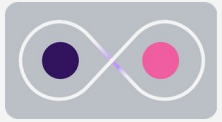
Qiskit Fall Fest 2026

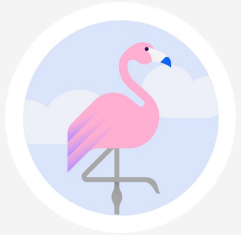
Qiskit

101

Manish B.







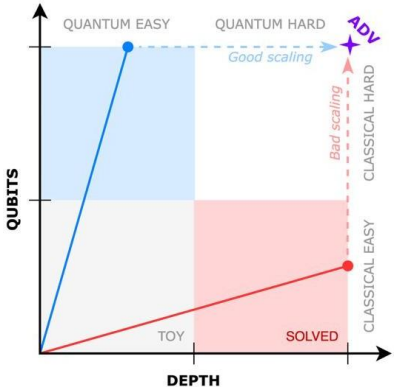
IBM

Overview

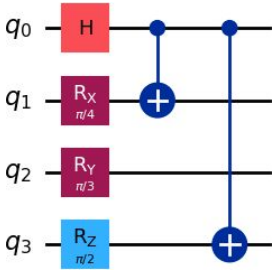
The Bloch sphere: a handy way to represent an arbitrary quantum state



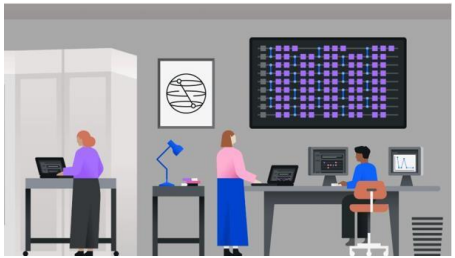
Shallow depth circuits and their utility for quantum



Universal quantum gate set: Single and two qubit gates



Hands-on quantum programming



What is a Quantum Computer ?

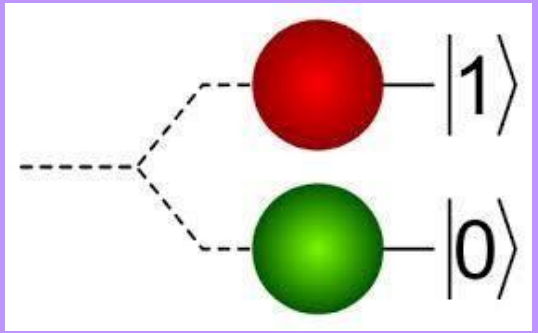
- Fundamental Building Blocks contains Qubits
- Computation is done through Applying Quantum Gates



What is a Qubit ?

- Any two-level Quantum System can be considered as qubits

- Ex: Two level atoms, Polarization of photons, Spins of electrons



Mathematical Representation of a Qubit

$$\begin{pmatrix}1\\0\end{pmatrix} = |0\rangle \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix} = |1\rangle,$$

- It is denoted by 2-orthonormal vectors, but a state of qubit can be in superposition these two states

$$|\varphi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

Bloch sphere: A compact representation of a quantum state

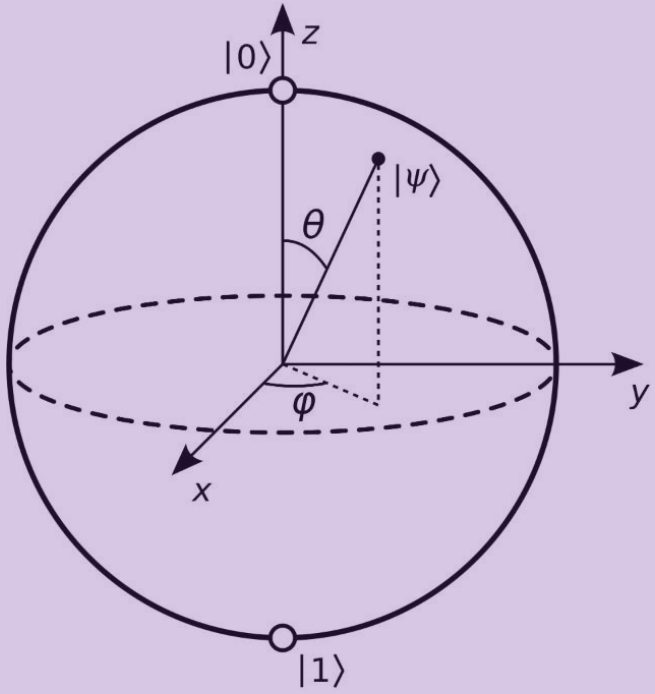
An arbitrary 2-state system (qubit) defined over the
complex numbers has 4 degrees of freedom.

Normalization and modding out by the overall phase reduces this to 2, which is why an arbitrary quantum state can be represented on a unit 2-sphere (called the “Bloch” sphere).

An arbitrary operation on a quantum state can be represented as a rotation on the Bloch sphere.

Pauli-X, Pauli-Y, Pauli-Z are generators of these rotations

$$|\psi\rangle = \cos\tfrac{\theta}{2}|0\rangle + e^{i\phi}\sin\tfrac{\theta}{2}|1\rangle$$



Quantum Gates



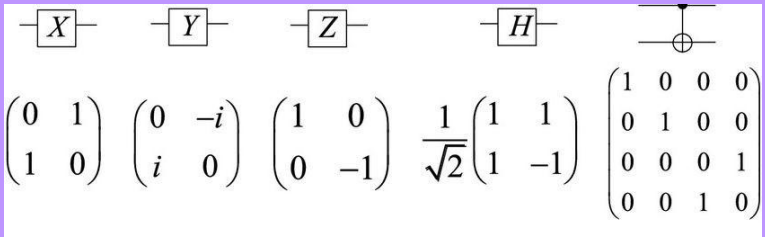
- We can manipulate qubits using quantum gates.

- Quantum Gates map qubits vectors to other qubit vectors thereby giving us a way to perform computation them.

Mathematical Representation of Quantum Gates

- Quantum Gates are represented by square matrices which act on the qubits vectors.
- Gates can be considered as rotations on the Bloch sphere

Mathematical Representation of Quantum Gates



$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

- The action of all the gates can be done by multiplying the matrix on the vector.
- We can have gates acting on single qubits as well as 2 qubits.

Single qubit gates



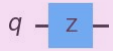
X (Bit flip) gate

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



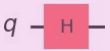
Y-gate

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$



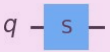
Z (Phase flip) gate

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



S-gate

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$



T-gate

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

Mathematical Representation of Many-qubits

- Many-qubit states are represented by taking tensor products of single qubit vectors

The tensor product of $|0\rangle$ and $|0\rangle$ to generate $|00\rangle$ is:

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

The tensor product of $|0\rangle$ and $|1\rangle$ to generate $|01\rangle$ is:

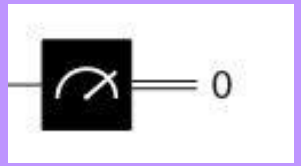
$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

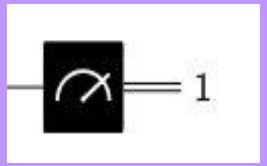
Similarly you can find the vector representation of $|10\rangle$ and $|11\rangle$

Quantum Measurement

- Measuring a quantum state collapses the wavefunction to either |0> or the |1> state.

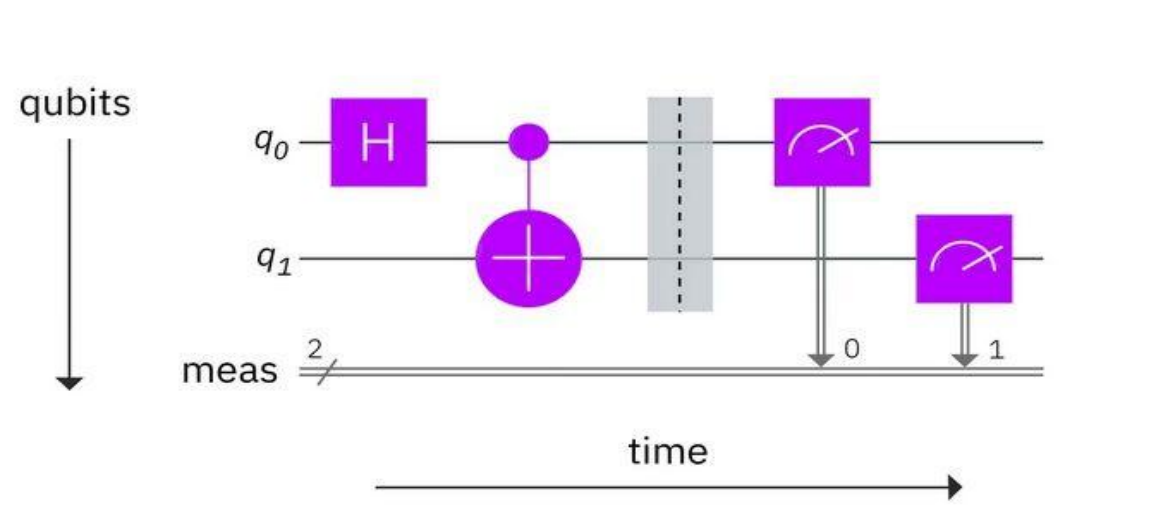
- The probability distribution would depend upon the coefficients - theta and phi



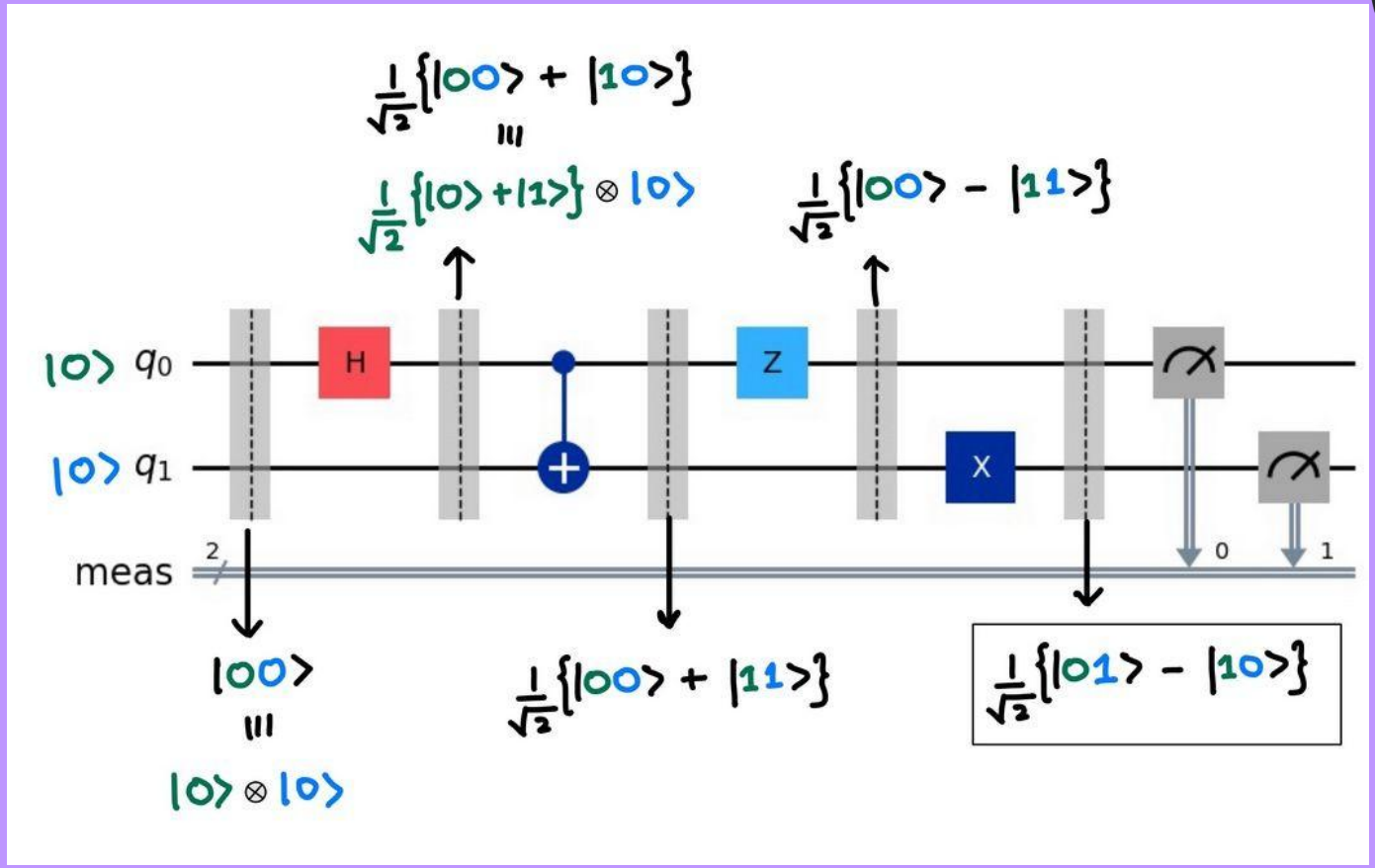


Quantum Circuits

- Combining all elements we have learnt till now we get what is called us a Quantum Circuit



Quantum Circuits



Now, On to Hands-On Lab!

